

An FFT Accelerator IP Core Implemented on FPGA

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Abstract—The main objective of this project is to design and implement a Fast Fourier Transform (FFT) IP core. Due to its ability to provide accurate frequency domain analysis of time domain signals, the FFT technique is indispensable in many applications related to digital signal processing. The implementation is based on the radix-2 method, whose lower computational complexity and hardware resource efficiency make it a good fit for FPGA platforms. With its robust FPGA and ARM CortexA53 processors, the ZCU104 board provides an adaptable environment for creating unique IP cores and incorporating them into more complex systems. The objective of this project is to use the Zynq SoC's hardware acceleration capabilities to improve the FFT core for speed and resource consumption. The ZCU104 board is used to test and validate the built IP core, proving its usefulness for real-time signal processing applications. As compared to software-based FFT implementations, the results demonstrate a notable boost in performance, underscoring the potential of FPGA based solutions in high performance computing jobs.

Keywords—Radix 2 FFT , Verilog , ZYNQ Soc ZCU104 board , DDR4 Memory , AXI4 bus , FPGA Signal Processing.

INTRODUCTION

In digital signal processing , the FFT technique is crucial because it makes it possible to quickly calculate the DFT of a sequence. In many fields, including audio signal processing, picture processing, telecommunications, and radar systems, where it is essential to convert time-domain signals into frequency-domain representations, the FFT is employed. The FFT is very useful in reducing the amount of operations needed, especially when utilizing the radix-2 approach, especially considering the computational intensity of the DFT.

High-performance FFT IP cores can be implemented in an optimal environment with the ZYNQ SoC ZCU104 FPGA board, an solid foundation that combines programmable logic and an ARM processing engine. Because of the board's architecture, embedded processors' software control and versatility can be paired with FPGAs' parallel processing capabilities. Our goal in this project is to balance speed, accuracy, and resource usage by creating a

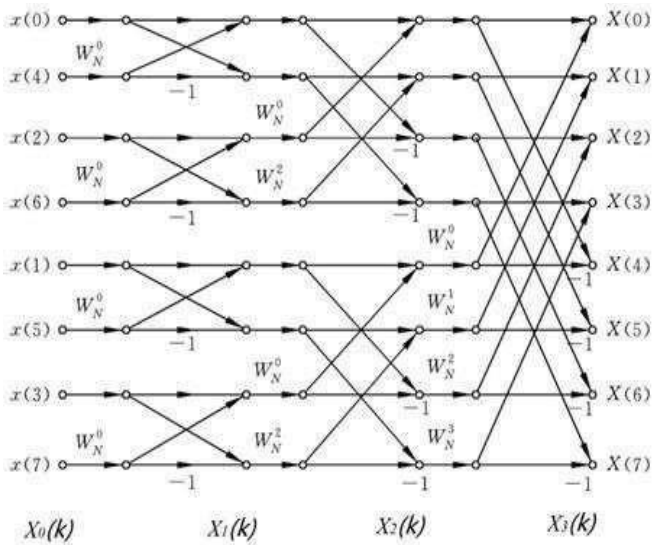
radix-2 FFT IP core specifically designed for the ZCU104 FPGA and taking advantage of its advantages. Radix-2 is a well-suited technique for FPGA implementation because of its efficiency and simplicity in the FFT algorithm. The FFT IP core must be designed, integrated with the ZYNQ SoC, and tested on hardware and in simulation to verify its functionality. The final objective is to create an FFT IP core that is highly efficient and flexible, suitable for a wide range of DSP applications, especially those that need real-time processing capabilities. With an emphasis on the ZCU104 board, this study adds to the continuing efforts to optimize FFT implementations on FPGA systems.

METHODOLOGY

This project's technique includes using the ZYNQ SoC ZCU104 FPGA board to develop, implement, and validate an FFT IP core. System design, IP core development, ZYNQ SoC integration, and hardware and simulation testing validation comprise the project's main stages. The FFT IP core's specifications and requirements are established during the first phase, which entails a thorough system design. Input data types, FFT sizes, and anticipated performance measures including speed, accuracy, and resource usage are all part of this process. Incorporating the ARM processing system and the programmable logic on the ZCU104 board, the system design also takes into account the integration of the IP core with the ZYNQ SoC. VHDL/Verilog is used for the hardware description during the construction of the FFT IP core. Because of its effectiveness and simplicity, the radix-2 algorithm is selected to provide an efficient design that can be implemented on the FPGA. The core's adjustable design allows it to accommodate a range of input sizes (64, 128, 256 points) and data formats (fixed-point, floating-point), allowing it to be tailored to the specific needs of different applications. Ensuring the butterfly calculation units, putting the twiddle factor multiplications into practice, and making sure that the data flows correctly through the FFT's steps are all part of the design process. Particular focus is placed on throughput maximization and delay minimization, which are essential for real-time signal processing applications. Through a recursive breakdown of the transform into smaller , the FFT Radix-2 algorithm used in this project computes Fast Fourier Transforms (FFTs) accurately and

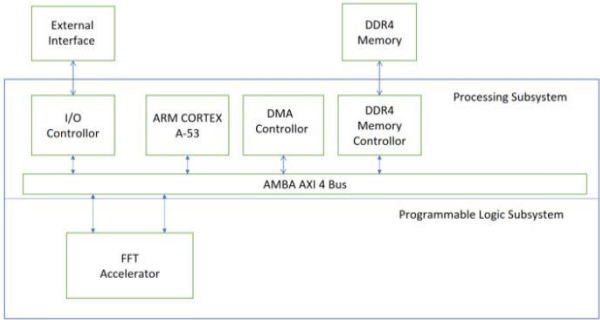
efficiently. To compute complex frequency-domain representations, it employs a divide and conquer strategy, progressively processing the input data. Radix-2 is a versatile tool for signal processing applications because it takes advantage of the FFT's intrinsic symmetry and periodicity qualities to compute FFTs effectively for input sizes that are powers of two. Fast and precise frequency analysis of input signals is made possible by this technique, which guarantees high-performance FFT computations within the FPGA-based FFT accelerator IP core.

Radix 2 FFT diagram shown below:



Hardware Implementation: The Zynq consist of quad-core ARM Cortex-A53 processing unit. In this FFT accelerator project, the ZCU104 board serves as the hardware platform for implementing and testing your custom IP core.we will use the ZCU104 board's capabilities in this FFT accelerator project to implement and assess the functionality of your custom IP core. Through the utilization of the ZCU104 board's FPGA fabric, ARM CPU, DDR4 memory, and peripheral connections, we may create a comprehensive solution for expediting FFT computations and incorporating them into more extensive embedded systems or signal processing applications. Furthermore, the accessibility of Xilinx's development tools and resources speeds up the project's time to-market and eases the design process.

Here we can see the hardware implementation of the FFT IP CORE.

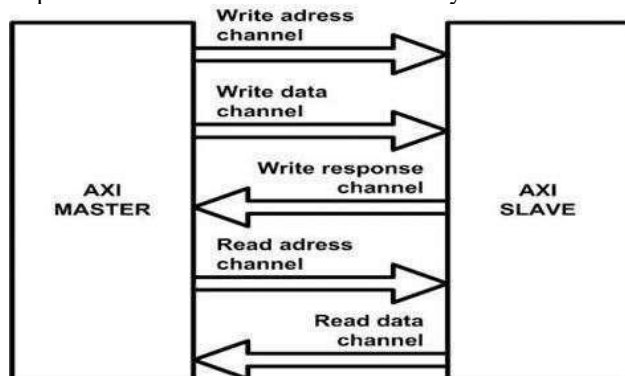


Here we can see the below table as when we perform the FFT on ARM and Accelerator separately, then we can see the ratio between them also:

No of points	FFT	On ARM	On FFT acceleration	Ratio
256	Radix 2	10 ³	10 ⁶	1000
512	Radix 2	10 ⁴	10 ⁶	100
1024	Radix 2	10 ⁵	10 ⁶	10
2048	Radix 2	10 ⁶	10 ⁶	1

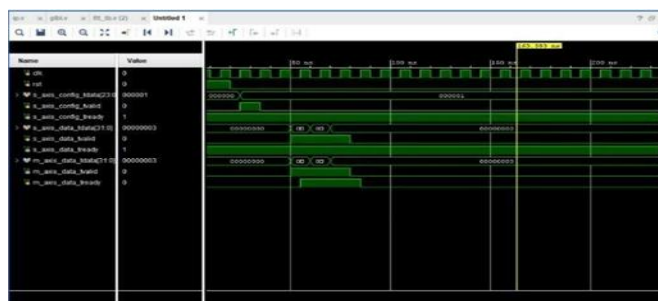
The ZYNQ SoC's Processing System (PS) is a dual-core Cortex-A53 processor with ARM architecture that manages software and control functions. It manages high-level operations like data management, system configuration, and connectivity with external peripherals and works in tandem with the FPGA fabric (Programmable Logic). In this project, the PS is in charge of starting and running the FFT IP core and overseeing data transfers via the AXI interface between memory and the FFT core. In order to facilitate integration with other software-based applications, it also processes the FFT findings. A balanced platform for real-time signal processing tasks is provided by the PS's flexibility and programmability, which enhance the programmable logic's high-speed operations.and also the FPGA part of the ZYNQ SoC that allows for the implementation of unique hardware designs is called Programmable Logic (PL). The FFT IP core in this project is implemented using the PL, which makes effective use of its parallel processing capabilities to carry out high speed computations.Hardware description languages (VHDL/Verilog) are used to set the PL. It can be customized to maximize throughput and minimize delay in FFT computations, for example. The PL is an essential part of high performance digital signal processing because its FFT core handles input signals in real-time and its flexibility permits future improvements.and the ZYNQ SoC uses DDR4 volatile memory to store data momentarily while it is

The ZYNQ SoC's AXI4 communication protocol is utilized to interface several parts, most notably the Processing System and the Programmable Logic. AXI4 is essential to this project since it connects the FFT IP core that was implemented in the PL to the rest of the system.



It makes high-speed data transfer possible, guaranteeing that the FFT core can efficiently handle incoming data from DDR4 memory and that the computed FFT results can be promptly returned to memory or processed further by the PS. Low-latency, high-throughput connection is guaranteed by AXI4, which is necessary to preserve the FFT IP core's real-time performance in signal processing applications.

HERE WE CAN SEE THE RESULT THAT IS IN THE FORM OF OUTPUT WAVEFORMS BY WRITING THE TESTBENCH VERILOG CODE FOR THE ABOVE FFT BLOCK DESIGN WHICH IS IMPLEMENTED IN VIVADO SOFTWARE.



In conclusion, the creation and application of the FFT accelerator IP core designed for FPGA-based systems—specifically, the Zynq SoC ZCU104 board—represent a

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